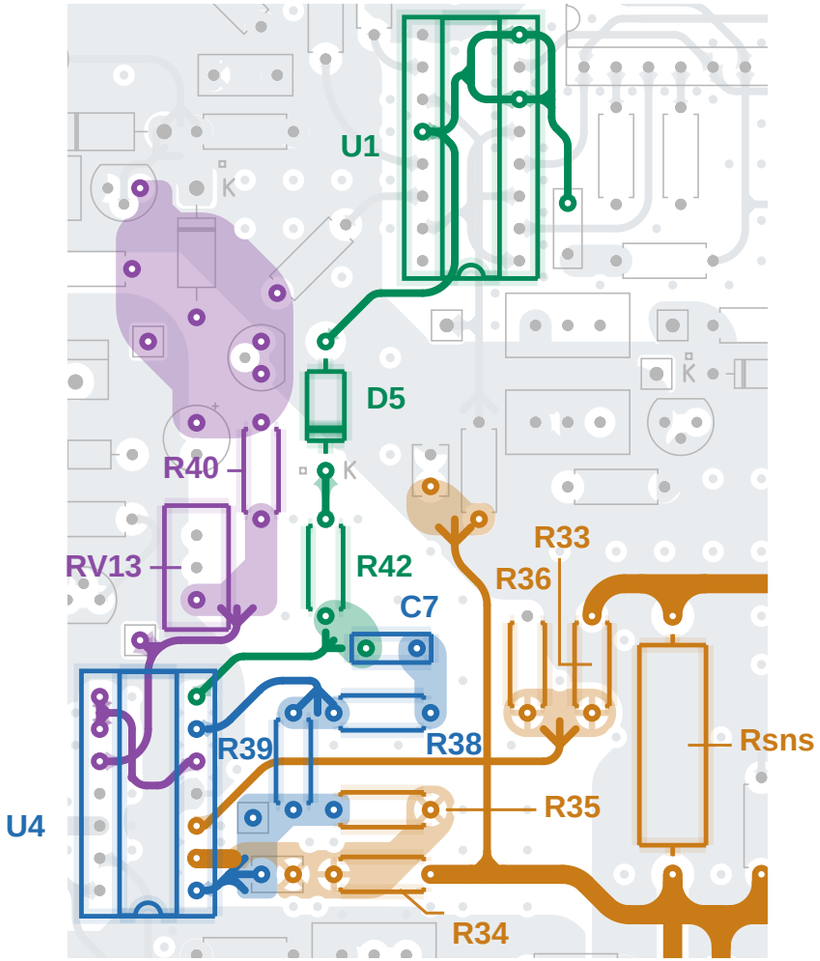
Current sensing and limiting

Actual KiCad copper + component highlights | Top-side projection of both outer layers



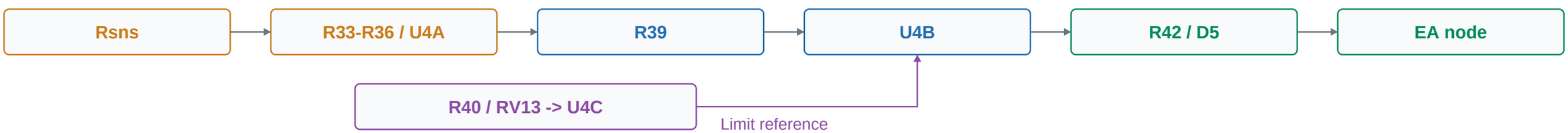
Differential sensing
R33 / R34 pick up opposite sides of Rsns. R35 / R36 complete the amplifier network.

Amplified signal
U4A output at pin 1 reaches U4B through R39. U4 contains several amplifier sections.

Adjustable limit
R40 / RV13 feed U4C pin 10. Its buffered threshold joins U4 pins 8, 9 and 5 via bottom copper and a via.

Limiting action
U4B output reaches the error-amplifier node through R42 and D5.

A shared color may span series components. Their pads and copper breaks remain distinct.



Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.